Code base

https://github.com/XiaohanZhangCMU/qwen-bpy-ft/tree/main

Collected data

data/processed/moonlake_sft.jsonl

- 125 trajectories.
- >2 turns 80
- >3 turns 45

Pipeline:

For each of 200+ scene seed descriptions, runs a repair loop:

1. Generate. Generator.complete() calls GPT-4o with a system prompt requiring valid bpy code
2. Execute. executor.py → shared/blender_runner.py:run_blender_script() runs the script headlessly in Blender and injects a manifest-collection snippet to capture scene state
3. Verify. scene_verifier.py:read_manifest() parses the JSON manifest (object/mesh/light counts etc)
4. Repair. if execution failed or scene too sparse, prompt_templates.py:format_repair_user_turn() feeds the stderr back to the model; repeats up to 4 times
5. Quality gate. quality_gate.py:check() accepts only if ALL 4 pass:
   - exit_code == 0
   - n_mesh ≥ 2
   - ≥ 2 turns total (forces multiple turns)
   - at least one repair turn (forces to have many examples with tool execution failures)

Checkpoints

Somehow the llama-factory works for qwen2.5 families but when I switch to qwen3.5 I hit IMA and didn't have enough time to figure out.

outputs/
qwen2_5_coder_3b_lora
qwen2_5_coder_7b_lora
Qwen3_5 → IMA

Evaluations

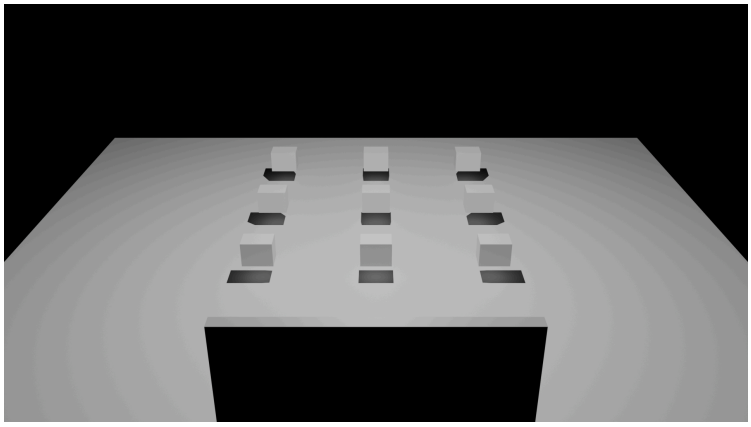
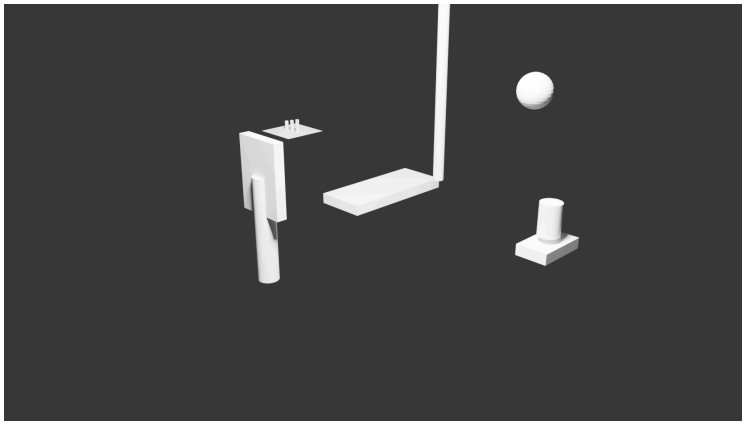
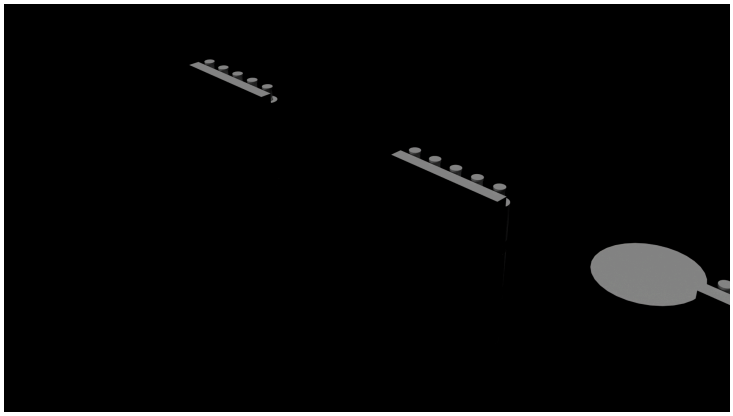
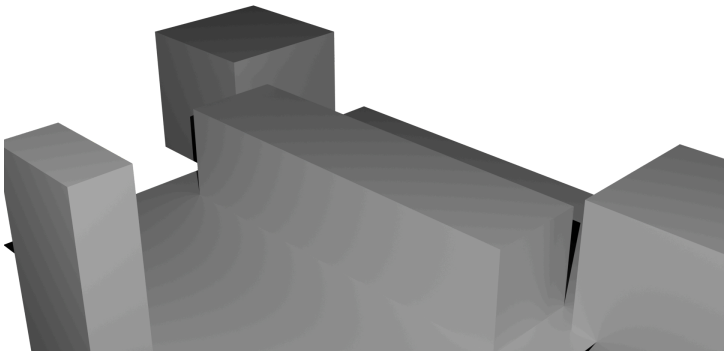
Vllm end points

```
-format` to override this.
(APIServer pid=2787550) INFO 03-31 00:14:49 [api_server.py:580] Starting vLLM server on http://0.0.0.0:8080
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:37] Available routes are:
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /openapi.json, Methods: GET, HEAD
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /docs, Methods: GET, HEAD
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /docs/oauth2-redirect, Methods: GET, HEAD
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /redoc, Methods: GET, HEAD
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /tokenize, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /detokenize, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /load, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /version, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /health, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /metrics, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/models, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /ping, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /ping, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /invocations, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/chat/completions, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/responses, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/responses/{response_id}, Methods: GET
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/responses/{response_id}/cancel, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/completions, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/messages, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/messages/count_tokens, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /inference/v1/generate, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /scale_elastic_ep, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /is_scaling_elastic_ep, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/chat/completions/render, Methods: POST
(APIServer pid=2787550) INFO 03-31 00:14:49 [launcher.py:46] Route: /v1/completions/render, Methods: POST
```

=== Quality metrics ===

Metric	Base Qwen-7B	**FT Qwen7B**	FT Qwen3B	GPT-4o
pass@1	33.3%	**40.0%**	20.0%	100.0%*
exec success rate	33.3%	**40.0%**	20.0%	100.0%*
mean objects/scene	3.3	**7.6**	2.0	21.7*

* = best in row

Vibe Checking Rendered Scene

Dentist office. Wine cellar. Class room….